

OCR Further Mathematics
Mechanics - Center of Mass
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
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Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		$\bar{x} = \frac{12a \times M + x \times m}{M + m} = \frac{12Ma + mx}{M + m}$	B1 (AO1.1) [1]	AG. www	
	b		$\bar{y} = \frac{3a \times M + y \times m}{M + m} = \frac{3Ma + my}{M + m}$	B1 (AO1.1) [1]		
	c		If P is at O, $\bar{x} = \frac{12Ma}{M + m}$ and $\bar{y} = \frac{3Ma}{M + m}$ $\bar{y} < 2a \Rightarrow 3M < 2M + 2m \Rightarrow$ $m > \frac{1}{2} M$ $\bar{x} < 6a \Rightarrow 12M < 6M + 6m \Rightarrow$ $m > M$ Conclusion: $m > \frac{1}{2} M$	B1ft (AO3.3) M1 (AO3.4) M1 (AO3.4) A1 (AO2.4) [4]	FT their expression for $\bar{y}$ AG.	Alternative: B1 for correct expressions for $\bar{x}, \bar{y}$ M1: forming 2 inequalities with 2a and 6a (must be right way around) M1: simplifying or manipulating both inequalities so that they can be combined or compared A1: fully correct and conclusion www
	d		$\bar{x} = \frac{12Ma + m \times 12ak}{M + m}$ used $\frac{12Ma + m \times 12ak}{M + m} = 6a$ $k = \frac{m - M}{2m}$ oe	B1 (AO3.3) M1 (AO3.4) A1 (AO1.1) [3]	Their $\bar{x}$ equated to 6a $k = \frac{1}{2} \left(1 - \frac{M}{m} \right)$	Ignore working with $\bar{y}$ Ignore working with $\bar{y}$ unless this affects final answer



Question			Answer/Indicative content	Marks	Part marks and guidance	
	e		$m = \frac{3}{2}M \Rightarrow k_{OC} = \frac{1}{6}$ $\bar{y} = \frac{3Ma + \frac{5}{2}M \times 6ak}{M + \frac{3}{2}M}$ $\bar{y} = 2a \Rightarrow \frac{6a + 18ak}{5} = 2a \Rightarrow k_{OA} = \frac{2}{9}$ (k changes from 1 to 0 and $k_{OA} > k_{OC}$ so) lamina topples over edge OA.	B1 (A03.3) M1 (A03.4) A1 (A03.4) A1 (A02.2a) [4]	$k_{OC} = \frac{3}{18} = 0.1\dot{6}$ Substituting $y = 6ak$ an $m = \frac{3}{2}M$ d into their $\bar{y}$ $k_{OA} = \frac{4}{18} = 0.\dot{2}$ www	
			Total	13		



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Question			Answer/Indicative content	Marks	Part marks and guidance		
2	a		The central radius is a line of symmetry of the shape.	B1 (AO2.4) [1]	Allow equal area each side		
	b		Area / mass of sector $\frac{1}{2}r^2 \times 2\theta$ and CoM of $\frac{2r \sin \theta}{3\theta}$ from centre of sector used at Area / mass of $(-)\frac{1}{2}r^2 \sin(2\theta)$ triangle and CoM of $\frac{2}{3}r \cos \theta$ triangle at from centre used $\frac{\frac{1}{2}r^2 \times 2\theta \times \frac{2r \sin \theta}{3\theta} - \frac{1}{2}r^2 \sin 2\theta \times \frac{2}{3}r \cos \theta}{\frac{1}{2}r^2 \times 2\theta - \frac{1}{2}r^2 \sin 2\theta}$ $= \frac{4r \sin \theta - 2r \sin 2\theta \cos \theta}{3(2\theta - \sin 2\theta)}$ $= \frac{2r(\sin \theta - \sin \theta \cos^2 \theta)}{3(\theta - \sin \theta \cos \theta)} = \frac{2r \sin \theta (1 - \cos^2 \theta)}{3(\theta - \sin \theta \cos \theta)}$ $= \frac{2r \sin^3 \theta}{3(\theta - \sin \theta \cos \theta)}$	B1 (AO1.2) B1 (AO1.2) M1 (AO3.1b) A1 (AO1.1) [4]	Must be used Using $\bar{x} = \frac{\sum m_i x_i}{\sum m_i}$ with sector and triangle of -ve mass, oe AG At least one intermediate step must be seen.	Must see change from double angle	
	c		Sector angle $2 \cos^{-1} \frac{2}{3}$ is $\theta = \cos^{-1} \frac{2}{3}$ or Area / Mass of component: $\pi \times 5^2 - \frac{1}{2} \times 3^2 (1.682 - \sin 1.682)$ $= 75.44$ $\bar{y} = \frac{(-3.097... \times 2.406...)}{75.44...}$ $= -0.0988$ (3 sf)	B1 (AO2.2a) M1 (AO1.1) A1 (AO1.1) M1 (AO3.1b) A1 (AO1.1) [5]	1.682... Attempting to find mass of component using 'negative' mass Their 3.097, 2.406 and 75.44	0.841... Allow use of 0.841 If $\theta = 2 \cos^{-1} \frac{2}{3}$ used this gives 1.095... instead of 2.406...	



Question			Answer/Indicative content	Marks	Part marks and guidance	
	d		$-0.0988M + 5m = 0$ So the mass of the component is $50.6m$ kg	M1 (A01.1) A1 (A01.1) [2]	May see Mg and mg	
			Total	12		



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Question			Answer/Indicative content	Marks	Part marks and guidance		
3	a		$\bar{x} = \frac{\int_0^5 x(6 + \sin x) dx}{\int_0^5 (6 + \sin x) dx} = \frac{72.622...}{30.716...}$ $= \frac{72.622...}{30.716...} = 2.36 \text{ (3 sf)}$	M1 (AO1.1) A1 (AO1.1) [2]	BC Attempt to use formula and either top or bottom correct soi AG. Both must be seen, or correct 2.364... seen		
	b		$\bar{y} = \frac{\int_0^5 \frac{1}{2}(6 + \sin x)^2 dx}{\int_0^5 (6 + \sin x) dx} = \frac{95.616...}{30.716...}$ 3.11 (3 sf)	M1 (AO1.1) A1 (AO1.1) [2]	BC Attempt to use formula and either top or bottom correct soi		
	c		The (part of the) binding (attached to the cover) is light oe The CoM of the badge is at A oe	B1 (AO3.5b) B1 (AO3.5b) [2]	eg The binding has no mass or the binding is very small so that the mass is concentrated at the hinge or the binding is smooth eg The badge is modelled as a particle or the badge is uniform		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
	d		$6 \times 2.36 \cos \frac{\pi}{3} + 2 \times 3 \cos \frac{\pi}{3}$ $= P \times 5$ $P = 2.02$	M1 (AO3.4) M1 (AO3.4) A1 (AO1.1) [3]	Total 'clockwise' moment about binding axis (allow inclusion of g if consist ent)... ...equals 'a nticlockwis e' moment	May use new $\bar{x} =$ 2.523... $8 \cos \frac{\pi}{3} \times \bar{x}$ = 5P	
			Total	9			



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Question			Answer/Indicative content	Marks	Part marks and guidance		
4	a		(G lies on the axis of) symmetry (of R)	B1(A01.2) [1]	Accept 'symmetrical'	Not 'distributed uniformly around the axis'	
					<u>Examiner's Comments</u> Most candidates answered this correctly. To achieve this mark examiners wanted to see that the candidate understood that the centre of mass lies on the line of symmetry. Common mistakes were reiterating the information that was given in the question (rotated about the x-axis) or by saying it was symmetrical about the y-axis.  Misconception Candidates referenced uniform distribution (which suggests a cylindrical shape)		



Question		Answer/Indicative content	Marks	Part marks and guidance		
	b	$\int_1^4 (2x^3 - 15x^2 + 36x - 20) dx = \frac{45}{2}$ $\int_1^4 (2x^4 - 15x^3 + 36x^2 - 20x) dx = \frac{1179}{20}$ $\bar{x} = \frac{\int_1^4 (2x^4 - 15x^3 + 36x^2 - 20x) dx}{\int_1^4 (2x^3 - 15x^2 + 36x - 20) dx}$ $\bar{x} = \frac{\frac{1179}{20}}{\frac{45}{2}} = \frac{131}{50} \text{ or } 2.62$	B1(A01.1a) B1(A01.1) M1(A01.1) A1(A01.1) [4]	BC With or without π soi BC With or without π soi Either no π or seen on both top and bottom	If treats as lamina award 0	

Examiner's Comments

Most candidates knew the formula here and were able to apply it effectively. If candidates didn't know the formula they were unable to access the question. Examiners expected to see calculator use to find the integrals involved. Some students spent far too long working out the integrations, the direction 'Find' means that they do not have to show detailed working.



Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		$PC = \sqrt{12} = 2\sqrt{3} = 3.46 \dots$ $\tan \theta = \frac{\sqrt{12}}{4 - 2.62} = (2.51 \dots)$ 1.19 rad or 68.3°	B1(A01.1) M1(A01.1) A1(A01.1) [3]	soi Allo $\frac{4 - 2.62}{\sqrt{12}}$ w (=0.398...)	C is the centre of the larger circle Can use a clearly stated value of PC $1 < 2.62 < 4$	
			Total	8			

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Question			Answer/Indicative content	Marks	Part marks and guidance		
5		a	Use $\bar{x} = \frac{\sum mx}{\sum m}$ or $\bar{y} = \frac{\sum my}{\sum m}$ of $\bar{x} = \frac{2 \times 0 + 3 \times 0.6 + 5 \times 0.4}{2 + 3 + 5} = \frac{3.8}{10} = 0.38$ (AG) $\bar{y} = \frac{2 \times 0 + 3 \times 0 + 5 \times 0.2}{2 + 3 + 5} = \frac{1}{10} = 0.1$	M1 (AO 1.2) A1 (AO 3.3) A1 (AO 3.3) [3]	Values must be substituted Intermediate step must be seen		
		b	G is not on the frame so the weight would have an unbalanced moment about the support point	E1 (AO 2.4) [1]			
		c	Moments about B: $0.6T = 10g \times 0.1$ $T = \frac{g(0.8-l)}{l}$ Eliminate $T: \frac{5g}{3} = \frac{g(0.8-l)}{l} \Rightarrow 8l = 2.4 \Rightarrow l = 0.3$ (AG)	M1 (AO 3.4) M1 (AO 3.4) A1 (AO 1.1) [2]	Each term must be of the form $F \times d$ Correct form for Hooke's law Intermediate working must be seen		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	Any two resolving equations oe, eg: (vertically): $R \cos \theta + F \sin \theta = 10g$ (horizontally): $R \sin \theta = F \cos \theta + T$ (parallel to plane): $F \cos \theta + T = 10g \sin \theta$ (perpendicular to plane): $R = T \sin \theta + 10g \cos \theta$ Use $T = \frac{5}{3}g$, $\sin \theta = \frac{3}{5}$, $\sin \theta = \frac{3}{5}$ to derive equations $4R + 3F = 50g$ and $9R = 12F + 25g$ oe $R = 9g$ and $F = \frac{14}{3}g$ $F \leq \mu R \Rightarrow \frac{14}{3}g \leq \mu \times 9g$ $\therefore \mu \geq \frac{14g}{3 \times 9g} = \frac{14}{27}$ (AG)	M1 (AO 3.4) M1 (AO 3.4) M1 (AO 1.1a) A1 (AO 1.1) M1 (AO 3.4) A1 (AO 1.1) [6]	Or taking moments eg about the point where the string is attached to the plane $(1 \times N = (0.8 + 0.1) \times 10g)$ or the point where the lines of action of R and the weight intersect: $\frac{0.1}{\sin \theta} \times F = \left(0.6 - \frac{0.1}{\tan \theta}\right) \times T$ Forming equations that lead to numerical values for F and R Both $F \leq \mu R$ with values substituted in Intermediate working must be seen	R is the normal contact force, F is the frictional force (up the slope)
			Total	13		



Question			Answer/Indicative content	Marks	Part marks and guidance	
6		a	OV is an axis of symmetry of S	E1 (AO 2.4) [1]		
		b	$x_H = \frac{3}{8} \times 0.4$ and $x_C = \frac{1}{4} \times 1.2$ Use of $\sum m_i x_i = (\sum m_i) \times OG$, or equivalent equation for moments about V (e.g.) $\frac{1}{3}\pi \times 0.4^2 \times 1.2 \times 0.3 + \frac{2}{3}\pi \times 0.4^3 \times (-0.15)$ $= \left(\frac{1}{3}\pi \times 0.4^2 \times 1.2 + \frac{2}{3}\pi \times 0.4^3\right) \times OG$ $\frac{8}{625} = \frac{8}{75} OG \Rightarrow OG = 0.12 \text{ m}$	B1 (AO 2.2a) M1 (AO 3.1b) A1 (AO 1.1) A1 (AO 1.1) A1 (AO 3.2a) [5]	$x_H = 0.15$ and $x_C = 0.3$ For the sum of the moments about O, one of the distances must be negative (or equivalent, e.g. difference of moments considered); for moments about V (e.g.) distances from V must be attempted AG so some intermediate working must be seen	One may be negative Masses may be represented by volumes; density may be present, but condone its absence



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	Moments about O (e.g.): $W \times 0.12 = T_V \times 1.2$ $T_V + T_O = W$ $T_O = 0.9W$ and $T_V = 0.1W$	M1* (AO 3.1b) dep* (AO 1.1) M1 A1 (AO 1.1) [3]	Taking moments: 2 terms if about O, G or V; 3 terms if about any other point Or a second moments equation, e.g. $W \times 1.08 = T_O \times 1.2$ Both	Each term must be of the form force $\times$ distance	
			Total	9			



Question			Answer/Indicative content	Marks	Part marks and guidance	
7		a	1	B1(A02.2a) [1]		
		b	$a \int_0^2 x(2-x) dx$ $\frac{4}{3}a$ $\frac{4}{3}a\bar{y} = a^2 \int_0^2 \frac{1}{2}x^2(2-x)^2 dx$ $= \frac{8}{15}a^2$ $\bar{y} = \frac{\frac{8}{15}a^2}{\frac{4}{3}a} = \frac{2}{5}a$	M1(A01.1a) A1(A01.1) M1(A01.2) A1(A01.1) A1(A02.2a) [5]	Attempt to integrate to find area. BC Use of BC Integral correctly evaluated AG	Ignore wrong/missing limits for M mark Ignore wrong/missing limits for M mark
		c	$3M\bar{Y} = 2M \times \frac{1}{2}a + M \times \left(a + \frac{2}{5}a\right)$ $\bar{Y} = \frac{4}{5}a$ $\tan 20^\circ = \frac{1}{\frac{4}{5}a}$ So value of a is 3.43 (3 sf)	M1(A03.1b) A1(A01.1) M1(A01.1) A1(A03.2a) [4]	Moments about base oe $\bar{Y} = -\frac{1}{5}a$ (eg wrt original x-axis) Use of CofM position to find a	or other horizontal axis
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
8		i	CoM of semi-circular lamina $\frac{4r}{3\pi}$ of radius r is $2r(10)\left(\frac{10}{2}\right) + \frac{1}{2}\pi r^2\left(10 + \frac{4r}{3\pi}\right) = \dots$ $\dots = \left(2r(10) + \frac{1}{2}\pi r^2\right)\bar{x}$ $\bar{x} = \frac{4r^2 + 30\pi r + 600}{3(40 + \pi r)}$	B1 M1 A1 A1 A1 [5]	oe, may be unsimplified Table of values idea AG Correctly shown, www. Examiner's Comments This part was often well done. The area of a semicircle was sometimes taken to be πr^2 rather than $\frac{1}{2}\pi r^2$ but most candidates were able to apply the formula to achieve the correct result for the CoM of a semi-circular lamina. Marks sometimes lost due to failure to add 10 for the CoM of the semi-circular lamina from AE. Candidates generally recognised the fact that 'show that' means detail is required.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\tan \theta = \frac{r}{10 - \bar{x}}$ $r \left(\frac{120 + 3\pi r}{1200 + 30\pi r - 600 - 30\pi r - 4r^2} \right) = \frac{1}{4}$ $(3\pi + 1)r^2 + 120r - 150 = 0$ $r = 1.14 \text{ only}$	M1 A1 A1 M1 A1 [5]	Using tan to find a relevant angle; accept reciprocal oe AG Correctly shown Attempt to solve quadratic in r 1.1375787... Examiner's Comments This part was much less successful, with many problems down to poor algebra. The correct angle was not always recognised, and a failure to subtract the CoM found in part (i) from 10 were the most common mistakes. The algebraic manipulation was beyond some candidates. Those who fared best had often rearranged to $x + 4r = 10$ before substituting in for x . Many candidates just picked up the final 2 marks for solving but there were also many candidates who omitted this part.	
			Total	10		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
9		i	$V_1 = \pi \int x^2 dy = \pi \int_0^{\ln 2} (e^{2y})^2 dy \text{ or } V = \pi \int_0^{\ln 2} (4^2 - (e^{2y})^2) dy$ $= \pi \left[\frac{1}{4} e^{4y} \right]_0^{\ln 2}$ $= \frac{\pi}{4} (e^{4 \ln 2} - 1) = \frac{15\pi}{4}$ $V = \pi (4)^2 (\ln 2) - \frac{15\pi}{4} = \frac{1}{4} \pi (64 \ln 2 - 15)$	M1 A1 A1 A1 [4]	For $\int (e^{2y})^2 dy$ For $\frac{1}{4} e^{4y}$ For correct substitution of limits AG www Examiner's Comments This question more than any highlighted the need for candidates to read the question carefully. Part (i), which was a given answer (as it was felt that this would benefit the candidates in answering the second part), clearly stated that the solid of revolution was formed by rotating the curve about the y-axis and so the volume should easily have been found by evaluating either $\pi \int_0^{\ln 2} (4^2 - (e^{2y})^2) dy$ or by considering $\pi (4)^2 (\ln 2) - \pi \int_0^{\ln 2} (e^{2y})^2 dy$. However, too many candidates found the volume of the solid formed when the curve was rotated about the x-axis and so spent many pages attempting to evaluate the incorrect integral	Limits not required for M and first A mark



Question			Answer/Indicative content	Marks	Part marks and guidance	
					$\pi \int_1^4 \left(\frac{1}{2} \ln x \right)^2 dx.$ A number of candidates surprisingly used a shell method to find the required volume and applied the less well known (but equally correct) formula $2\pi \int_a^b xf(x)dx.$ However, these candidates then had to spend a considerably amount of time using integration by parts when it came to evaluating the integral $2\pi \int_1^4 x \left(\frac{1}{2} \ln x \right) dx.$	
		ii	$V_1\bar{y} = \pi \int yx^2 dy = \pi \int_0^{\ln 2} y(e^{2y})^2 dy = \pi \int_0^{\ln 2} ye^{4y} dy$ $\text{or } V_1\bar{y} = \pi \int 16y - ye^{4y} dy$ $= \pi \left\{ \left[\frac{1}{4} ye^{4y} \right]_0^{\ln 2} - \frac{1}{4} \int_0^{\ln 2} e^{4y} dy \right\} = \pi \left[\frac{1}{4} ye^{4y} - \frac{1}{16} e^{4y} \right]_0^{\ln 2}$ $V_1\bar{y} = \pi \left(\ln 16 - \frac{15}{16} \right)$ $\left(\frac{1}{2} \ln 2 \right) \left(\pi (4)^2 (\ln 2) \right) - \pi \left(\ln 16 - \frac{15}{16} \right) = \bar{y} \left(\pi (4)^2 (\ln 2) - \frac{15\pi}{4} \right)$ $\bar{y} = \frac{128(\ln 2)^2 - 16 \ln 16 + 15}{256 \ln 2 - 60}$	*M1 M1dep* A2 M1 A1 A1	For $\int yx^2 dy$ <	



Question			Answer/Indicative content	Marks	Part marks and guidance		
				[7]	from (i)		
					Two terms correct	For	
					Exact (oe)	guidance:	
					Examiner's Comments Part (ii) was the least well answered part-question on the entire paper with many candidates not realising that the exact y-coordinate of the centre of mass of the solid was found by considering either $V\bar{y} = \pi \int_0^{\ln 2} 16y - ye^{4y} dy$ or by evaluating $\frac{\pi}{V} \int_1^4 x \left(\frac{1}{2} \ln x \right)^2 dx$ Those that did were usually successful in dealing with the required integration.		
			Total	11			



Question			Answer/Indicative content	Marks	Part marks and guidance	
10		i	$I = \frac{1}{3}(2M)(2a)^2 + 2M(ka)^2$ $T = 2\pi \sqrt{\frac{8Ma^2 + 6Mk^2a^2}{3(2Mg)(ka)}}$ $T = 2\pi \sqrt{\frac{a}{3g} \left(\frac{4 + 3k^2}{k} \right)}$	B1 B1 M1 A1 [4]	B1 for each term SC B1: for incorrect masses in both expression $T = 2\pi \sqrt{\frac{I}{mgh}}$ with correct mass AG www	Examiner's Comments Part (i) was answered extremely well with the majority of candidates correctly applying the standard formula for the period of small oscillations of a compound pendulum. The main issue came in the calculation of the moment of inertia about the horizontal axis when a number of candidates $2M(2a)^2$ to $\frac{1}{3}(2M)(2a)^2$. Furthermore, a number of candidates incorrectly added values for the mass and length of the rod.



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{d}{dk} \left(\frac{4+3k^2}{3k} \right) = 0$ $\frac{3k(6k) - (4+3k^2)(3)}{(3k)^2} = 0 \Rightarrow k^2 = \dots$ $k^2 = \frac{4}{3}$	M1 M1 A1 [3]	Attempt to differentiate $f(k)$ and set equal to zero Correct application of quotient rule (or other correct method) and get to $k^2 = \dots$ or $k = \dots$ (if k^2 not stated first) Cao (oe) Examiner's Comments In part (ii) the vast majority understood the need to differentiate, with respect to k, the expression $\frac{4+3k^2}{k}$ and to then set this answer equal to zero to find the value of k^2 for which the period of oscillations would be least. A number of candidates made this part significantly more demanding for themselves by attempting to differentiate the whole expression for T. Some candidates did not read this part carefully and gave the value for k without ever stating the value for k^2.	Or differentiate s wrt k eg $\frac{d}{dk} \left(\frac{4}{3k} + k \right) = 0$
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance		
11		i	$Pl\sqrt{5} = W \times 3l \cos \theta$ $P = 1.2W$ $Ql\sqrt{2} = U \times \frac{\lambda}{2}l \cos \emptyset$ $Q = 0.25\lambda U$	M1 A1 M1 A1 [4]	Mom about A for AB AG Mom about A for AC	Allow $\sin \theta$, cancelled / Not from use of angle 26.565° Allow $\sin \emptyset$, cancelled /	
					Examiner's Comments This part was done well by most candidates. However, some candidates worked out the angle at P in degrees, rather than using an exact value when taking moments, and so dropped a mark. Others made a slip and arrived at $\frac{1}{2}\lambda U$ for the force at Q.		
		ii	(H) $P \sin \theta = Q \sin \emptyset$ (V) $W + U = P \cos \theta + Q \cos \emptyset$ $W + U = P \cos \theta + P \sin \theta \times \frac{\cos \emptyset}{\sin \emptyset}$ $W + U = \frac{3}{\sqrt{5}} \times 1.2W$ $k = 0.610$	M1 M1 A1 M1* *M1 A1 A1	$P \frac{1}{\sqrt{5}} = Q \frac{1}{\sqrt{2}}$; compts essential $W + U = P \frac{2}{\sqrt{5}} + Q \frac{1}{\sqrt{2}}$; compts essential Eliminate Q (or P) dep M1M1 Elim P and Q to get equation in	Allow $\frac{2}{\sqrt{5}}$ for M1 Allow $\frac{1}{\sqrt{5}}$ for M1, sign errors $W + U = Q \cos \theta \times \frac{\sin \emptyset}{\sin \theta} + Q \cos \emptyset$ $[W + U = 0.25\lambda U \times \frac{3}{2\sqrt{2}}]$ 0.6099689 4.97695	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			$\lambda = 4.98$ $[P\sqrt{5}l - W3l \cos \theta = Q\sqrt{2}l - U\frac{\lambda}{2} \cos \phi]$	[7]	$\begin{array}{r} k, W + U = \\ 1.609689W \\ 18\sqrt{5} - 25 \\ \hline 25 \end{array}$ Mom about A (or any other point) for whole system – allow M1(A1) if resolving not seen twice] Allow use of angles in (ii): 26.6 & 45 OR after M1M1A0/1, M1* for 2 equns in terms of k and λ, *M1 for solving for k or λ.	
					Examiner's Comments This part was omitted by quite a number of candidates, and many others wrote down an equation or two, often correctly, but then made no further progress. There were also a good number of perfectly correct solutions; some of them were very efficient. This question needs to be tackled by resolving vertically and horizontally for the whole system, to get two equations in terms of P, Q, W and U, though many candidates used the results from part (i) to avoid using P and Q. P and Q can be eliminated, and U written as kW to give two equation in k and λ which can be solved simultaneously. Some candidates found an efficient short cut when they realised that the $Q \sin 45^\circ$ term in one equation was equal to the $Q \cos 45^\circ$ term in the other. This led them	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					to find the value of k quickly but some then failed to find the value of λ. Taking moments about A for the whole system does not work – the terms cancel out since this has effectively been used to do part (i). Taking moments about B or C for the whole system is extremely complicated, because the directions of the forces at P and Q are difficult to manage. Working on either rod AB or rod AC means introducing components of the force at A; marks were only allocated to this approach when these components were eliminated.	
			Total	11		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
12		a	$A = \int_0^{\ln 32} e^{0.8x} dx = \frac{75}{4}$ $A\bar{x} = \int_0^{\ln 32} xe^{0.8x} dx$ $= \left[1.25xe^{0.8x} \right]_0^{\ln 32} - 1.25 \int_0^{\ln 32} e^{0.8x} dx$ $= \left[1.25xe^{0.8x} - 1.5625e^{0.8x} \right]_0^{\ln 32}$ $= (20 \ln 32 - 25) - (0 - 1.5625) \Rightarrow \bar{x} = \dots$ $\bar{x} = \frac{16}{3} \ln 2 - \frac{5}{4}$	M1(A01.2) A1(A01.1) M1(A03.1a) A1(A01.1) A1(A01.1) M1(A01.1) E1(A02.1) [7]	Attempt to integrate to find area BC Use integration by parts Correctly applied to end of first stage Using correct limits and $\bar{x} = \frac{A\bar{x}}{x}$ www; AG	Limits not required for M mark $\left[1.25e^{0.8x} \right]_0^{\ln 32}$	
		b	$A\bar{y} = \frac{1}{2} \int_0^{\ln 32} (e^{0.8x})^2 dx = \frac{1275}{16}$ $\bar{y} = \frac{17}{4}$	M1(A02.1) A1(A01.1) A1(A01.1) [3]	For $\dots$ $\frac{1}{2} \int y^2 dx$ and attempt to integrate BC	$\frac{1}{2} \left[\frac{e^{1.6x}}{1.6} \right]_0^{\ln 32}$	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$\bar{X}\left(16\ln 32 - \frac{75}{4}\right) = (8)(16)(\ln 32) - \left(\frac{17}{4} \times \frac{75}{4}\right)$ $\bar{X} = 9.92$	M1(A03.1a) M1(A02.1) A1FT(A01.1) A1(A01.1) [4]	Table of values idea to get an equation / expression Attempt at $cv(x) = (8)(16)(\ln 32) - cv(A\bar{x})$ 9.915779...		
			Total	14			



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Question			Answer/Indicative content	Marks	Part marks and guidance	
13		i	$BD = 2a \cos \theta$	B1	Award if seen in EPE term	
		i	$GPE \text{ for rod } AB = (-)mg \left(a + \frac{1}{2}a \cos 2\theta \right)$	B1	$(-)mg \left(a + \frac{1}{2}a \sin(90 - 2\theta) \right)$	
		i	$GPE \text{ for rod } BC = (-)mg \left(a + a \cos 2\theta - \frac{1}{2}a \sin 2\theta \right)$	B1	$(-)mg \left(a + a \sin(90 - 2\theta) - \frac{1}{2}a \cos(90 - 2\theta) \right)$	
		i	$EPE = \frac{\lambda mg (2a \cos \theta - a)^2}{2a}$	M1	Using $\frac{\lambda x^2}{2a}$ with their BD	
		i	$V = \frac{1}{2}mga(\sin 2\theta - 3 \cos 2\theta) + \frac{1}{2}\lambda mga(2 \cos \theta - 1)^2 - 2mga$	A1	AG correctly shown	
		ii	$\frac{dV}{d\theta} = \frac{1}{2}mga(2 \cos 2\theta + 6 \sin 2\theta) +$	M1	Attempt at differentiation	
		ii	$\lambda mga(2 \cos \theta - 1)(-2 \sin \theta)$	A1	Correct derivative ...+ $\lambda mga(2 \sin \theta - 2 \sin 2\theta)$	
		ii	$\frac{1}{2}(2(0)+6)+\lambda \left(\frac{2\sqrt{2}}{2}-1 \right) \left(-\frac{2\sqrt{2}}{2} \right) = 0$	M1	Set $\frac{dV}{d\theta} = 0$ and $\theta = \frac{1}{4}\pi$	
		ii	$\lambda = \frac{3}{2-\sqrt{2}}$	A1	oe eg $\frac{6+3\sqrt{2}}{2}$	
		iii	$\frac{d^2V}{d\theta^2} = \frac{1}{2}mga(-4 \sin 2\theta + 12 \cos 2\theta)$	M1	Attempt at second derivative	
		iii	$-2\lambda mga(2 \cos 2\theta - \cos \theta)$	A1	oe ...- $2\lambda mga[(2 \cos \theta - 1) \cos \theta - 2 \sin^2 \theta]$	
		iii		M1	Set $\theta = \frac{1}{4}\pi$ and using their λ (must be evidence of substitution)	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\frac{d^2V}{d\theta^2} = mga(1 + 3\sqrt{2}) > 0,$ so equilibrium is stable	A1	Correct value of V'' and > 0 $V'' = (5.24264\dots)mga$	Examiner's Comments The vast majority of candidates correctly showed the given result for the gravitational potential energy of the system in part (i). Nearly all candidates correctly calculated the length of BD as $2a\cos\theta$ although some spent a considerable amount of time using, and simplifying, their length for BD (after applying the cosine rule) rather than exploiting the fact that triangle ABD was isosceles. Those that did have the correct length for BD nearly always went on to correctly find the elastic potential energy of the string. A number of candidates struggled with finding the gravitational potential energy of the two rods AB and BC with a number not showing sufficient detail of how they arrived at the given answer. While nearly all candidates correctly understood the method for calculating the exact value of λ in part (ii) for some, however, the differentiation of V required in this part was far too demanding. Many candidates struggled to calculate $\int_0^2 2xe^{\frac{1}{2}x} dx$ accurately with many omitting the negative sign or failing to include an additional factor of 2. In part (iii) most candidates



Question			Answer/Indicative content	Marks	Part marks and guidance
					scored both method marks for attempting to differentiate their $\frac{dV}{d\theta}$ and substituting $\frac{1}{4}\pi$ for θ but very few correctly had the value of the second derivative as $mga(1+3\sqrt{2})$ and hence could accurately determine that the equilibrium position was stable.
			Total	13	



Question			Answer/Indicative content	Marks	Part marks and guidance		
14		i	$A_1 = \int_0^2 2e^{\frac{1}{2}x} dx$	M1*	Attempt at integration to find area	Limits not required for M mark	
		i	$= \left[4e^{\frac{1}{2}x} \right]_0^2 = 4(e - 1)$	A1			
		i	$A_1 \bar{y} = \frac{1}{2} \int_0^2 \left(2e^{\frac{1}{2}x} \right)^2 dx = \frac{1}{2} \int_0^2 4e^x dx$	M1*	Attempt at integration	Limits not required for M and A marks	
		i	$= \frac{1}{2} \left[4e^x \right]_0^2 (= 2e^2 - 2)$	A1			
		i	$\bar{y} = \frac{2e^2 - 2}{4(e - 1)} = \frac{e + 1}{2}$	M1 dep*	M1 for $\bar{y} = \frac{A\bar{y}}{A}$		
		i		A1			
		ii	$A_\Delta = \frac{1}{2}(3)(2e)$	B1	Integration by parts	Or by integration	
		ii	$A_2 \bar{x} = \int_0^2 x \left(2e^{\frac{1}{2}x} \right) dx$	M1*		Clear indication of integrating exponential terms and differentiating x term	
		ii	$= \left[4xe^{\frac{1}{2}x} \right]_0^2 - \int_0^2 4e^{\frac{1}{2}x} dx$				
		ii	$= \left[4xe^{\frac{1}{2}x} - 8e^{\frac{1}{2}x} \right]_0^2 (= 8)$	A1 A1			All terms integrated correctly (A1 for one error)
		ii	$(A_1 + A_\Delta) \bar{x} = cv(8) + 3A_\Delta$	M1 dep*			M1 for table of values idea
		ii	$\bar{x}(4e - 4 + 3e) = 8 + 3(3e)$	A1		Two terms correct	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\bar{x} = \frac{9e+8}{7e-4}$	A1	oe	Examiner's Comments In part (i) a small minority of candidates attempted to find the value of the x-coordinate of the correct lamina or instead found the y-coordinate of the uniform lamina given in part (ii). However, the majority of candidates correctly found the area of the lamina in part (i) but many did not go and correctly evaluate the integral $\frac{1}{2} \int y^2 dx$ with the most common error being to forget the factor of $\frac{1}{2}$. While the vast majority of candidates attempted to use strips parallel to the y-axis in their attempt to find the required centre of mass a number attempted to use strips parallel to the x-axis but none were successful due to the difficulty in dealing with the corresponding geometry of the problem. Many did not appreciate that the centre of mass in this case would be given by $\bar{y} = \frac{\int_{y=2}^{2e} y \left(2 - 2 \ln \left(\frac{y}{2} \right) \right) dy + 4}{4e - 4}$ as (in this case) the horizontal strips are formed from the line $x = 2$ rather than the y-axis. Part (ii) was answered extremely well with many obtaining the correct exact



Question			Answer/Indicative content	Marks	Part marks and guidance	
						value for the x- coordinate of the centre of mass of the lamina. Many dealt correctly with the required integration by parts for the integral $\int_0^2 2xe^{\frac{1}{2}x} dx$ but it was surprising how many candidates used integration to work out the centre of mass of the triangular part of the uniform lamina when it was expected that candidates would simply state that the centre of mass (of this triangular lamina) was at a horizontal distance of 3 units from the y- axis.
			Total	13		



Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	$A = \int_1^2 kx^2 dx = \left[\frac{kx^3}{3} \right]_1^2 = \frac{7k}{3}$	*M1 A1	M1 attempt to integrate to find area. Limits not required for M mark	
		i	$A\bar{x} = \int_1^2 x(kx^2) dx = \left[\frac{1}{4} kx^4 \right]_1^2 = \frac{15k}{4}$	*M1 A1	M1 for ... $\int x^3 dx$ and attempt to integrate. Limits not required for M mark	
		i	$\bar{x} = \left(\frac{15k}{4} \right) \left(\frac{3}{7k} \right)$	M1 dep*	M1 for $\bar{x} = \frac{A\bar{x}}{A}$	
		i	$\bar{x} = \frac{45}{28}$	A1	Examiner's Comments Most candidates correctly found the x - coordinate of the centre of mass of the lamina.	
		ii	$A\bar{y} = \frac{1}{2} \int_1^2 (kx^2)^2 dx = \frac{1}{2} k^2 \left[\frac{x^5}{5} \right]_1^2 = \frac{31k^2}{10}$	*M1 A1	M1 for ... $\frac{1}{2} \int y^2 dx$ and attempt to integrate	
		ii	$\left(\frac{31k^2}{10} \right) \left(\frac{3}{7k} \right) = \frac{45}{28}$	M1 dep*	$cv(\bar{x}) = cv(\bar{y})$ - this mark is dependent on scoring all M marks in (i) and (ii)	
		ii	$k = \frac{75}{62}$	A1	Cao	



Question			Answer/Indicative content	Marks	Part marks and guidance
		ii	Or $\bar{Ay} = \frac{1}{4\sqrt{k}} \int_k^{4k} y^{\frac{3}{2}} dy = \frac{1}{4\sqrt{k}} \left[\frac{2}{5} y^{\frac{5}{2}} \right]_k^{4k} = \frac{31k^2}{10}$		M1 for $\frac{1}{4\sqrt{k}} \int y^{\frac{3}{2}} dy$ and attempt to integrate Examiner's Comments The ability of the candidates to find the exact value of k was significantly more varied. It was expected that candidates would first find the y - coordinate of the centre of mass of the lamina by consider the integral $\frac{1}{2} \int_1^2 y^2 dx$ but it was relatively common for the factor of $\frac{1}{2}$ to be missing. A number of candidates tried to integrate about the y-axis but this was rarely successful as many stated that the required integral was $\frac{1}{\sqrt{k}} \int_k^{4k} y^{\frac{3}{2}} dy$, forgetting the factor of a $\frac{1}{4}$, which is required as $y = kx^2 \Rightarrow dy = 2kx dx$. Those candidates who did find the y - coordinate of the centre of mass correctly usually went on to find the correct value for k.
			Total	10	



Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	$HB = 2r \cos\theta$ and $HP = a - 2r \cos\theta$	B1B1		
		i	$V = -\lambda mg(HP) - mg(HB) \cos\theta$	M1	Attempt at V with their HP and HB	
		i	$V = -\lambda mg(a - 2r \cos\theta) - 2mgr \cos^2\theta$	A1	One term correct	
		i	$V = mg (2\lambda r \cos\theta - 2r \cos^2\theta - \lambda a)$	A1	AG www correctly obtained If M0 then SC B1 for one term correct	
					Examiner's Comments Most candidates correctly derived the given expression for the total potential energy with only a small minority having difficulty with calculating the distances HB and HP .	
		ii	$\frac{dV}{d\theta} = -2\lambda mgr \sin\theta + 4mgr \cos\theta \sin\theta = 0$	M1	Attempt at differentiation	
		ii	$2mgr \sin\theta(2 \cos\theta - \lambda) = 0$	A1	Correct derivative and equal to zero	
		ii	$\frac{1}{2} \leq \cos\theta < 1$	M1	Comparing their $\cos\theta < 1$ $\cos\theta \geq \frac{1}{2}$ - condone $\cos\theta \leq 1$	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\Rightarrow 1 \leq \lambda < 2$	A1	Examiner's Comments	
					Candidates found this the most challenging part of the paper with only a small minority scoring all four marks. While the majority of candidates obtained the two correct critical values of λ , only a small minority appreciated that $1 \leq \lambda < 2$; when $\lambda = 2$ the values of $\cos \theta$ and $\sin \theta$ are equal and therefore there would only be the one equilibrium position.	
		iii	$\frac{d^2V}{d\theta^2} = -2\left(\frac{3}{2}\right)mgr \cos \theta + 4mgr \cos 2\theta$	*M1 A1	M1 Attempt to differentiate V' (or first derivative test) – allow in terms of λ	
		iii		M1 dep*	M1 sub. their first angle into V'' (maybe implied by later working (eg correct sign and conclusion))	
		iii	$\sin \theta = 0 \Rightarrow V'' = mgr > 0 \therefore$ stable	A1	A1 correct (unsimplified) value of V'' and > 0	
		iii		M1 dep*	M1 sub. their second angle into V'' (maybe implied by later working (eg correct sign and conclusion))	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\cos \theta = \frac{3}{4} \Rightarrow V'' = -\frac{7}{4}mgr < 0 \therefore \text{unstable}$	A1	A1 correct (unsimplified) value of V'' and < 0 If both values of V'' correct and correct conclusion then award B1 if no consideration of sign seen Examiner's Comments Nearly all candidates stated that when the first derivative of V was zero then either $\sin \theta = 0 \text{ or } \cos \theta = \frac{3}{4}$ Most candidates correctly went on to find the second derivative of V and they knew that they had to find the sign of the second derivative for their values of θ. In general, the work for $\sin \theta = 0$ was very good, but many struggled to make any progress with $\cos \theta = \frac{3}{4}$ often because they did not rewrite their expression for the second derivative of V in terms of $\cos \theta$.	
			Total	15		



$= \left[\sqrt{2} \sin x + \frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{2}} = \frac{1}{2}$	A1 for $\frac{1}{2}$
Integration by parts	Clear indication of



Question			Answer/Indicative content	Marks	Part marks and guidance	
			OR $\frac{\int_0^{\frac{\pi}{4}} x \sin 2x dx}{\int_0^{\frac{\pi}{4}} \sin 2x dx} = \frac{\left[-\frac{x}{2} \cos 2x\right]_0^{\frac{\pi}{4}} + \frac{1}{2} \int_0^{\frac{\pi}{4}} \cos 2x dx}{\left[-\frac{1}{2} \cos 2x\right]_0^{\frac{\pi}{4}}}$ $= \frac{\left[-\frac{x}{2} \cos 2x + \frac{1}{4} \sin 2x\right]_0^{\frac{\pi}{4}}}{\frac{1}{2}} = \frac{1}{2}$ $\frac{\int_0^{\frac{\pi}{4}} x \sqrt{2} \cos x dx}{\int_0^{\frac{\pi}{4}} \sqrt{2} \cos x dx} = \frac{\sqrt{2} \left[x \sin x\right]_0^{\frac{\pi}{4}} - \sqrt{2} \int_0^{\frac{\pi}{4}} \sin x dx}{\left[\sqrt{2} \sin x\right]_0^{\frac{\pi}{4}}}$ $= \frac{\sqrt{2} \left[x \sin x + \cos x\right]_0^{\frac{\pi}{4}}}{1} = \sqrt{2} \left(\frac{\pi \sqrt{2}}{8} + \frac{\sqrt{2}}{2} - 1\right)$ $\left(1 - \frac{1}{2}\right)x = \left(\frac{\pi}{4} + 1 - \sqrt{2}\right)(1) - \frac{1}{2}\left(\frac{1}{2}\right)$ $\bar{x} = \frac{\pi}{2} + \frac{3}{2} - 2\sqrt{2}$	A1 M1* A1 A1 M1* A1 A1 M1 dep* A1	Attempt at integration to find the centre of mass for the lamina bounded by $y = \sin 2x$ A1 for correct first stage of parts in numerator and correct denominator (ignore limits for this mark) Cao Attempt at integration to find the centre of mass for the lamina bounded by $y = \sqrt{2} \cos x$ A1 for correct first stage of parts in numerator and correct denominator (ignore limits for this mark) Cao Taking moments – signs must be consistent	For the first 2 M marks must be a genuine attempt at integrating by parts
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
18		i	$GPE = -mg a \sin \theta$	B1		
		i	$x^2 = a^2 + a^2 - 2(a)(a)\cos \theta$	B1	Or $x = 2a \sin \left(\frac{\theta}{2} \right)$	
		i	$EPE = \frac{\sqrt{3}mg \{2a^2(1 - \cos \theta)\}}{2a}$	M1	Using $\frac{\lambda x^2}{2a}$	Genuine attempt at extension
		i	$V = mga(\sqrt{3} - \sin \theta - \sqrt{3} \cos \theta)$	A1	AG Correctly shown Examiner's Comments Most candidates derived the given expression for the total potential energy correctly with only a small number having difficulty with calculating the extension in the string. Establishing the position and stability of equilibrium	
		ii	$\frac{dV}{d\theta} = mga(-\cos \theta + \sqrt{3} \sin \theta) = 0$	M1	Attempt at differentiation	
		ii		A1	Correct derivative and equal to zero	
		ii	$\sqrt{3} \tan \theta - 1 = 0 \Rightarrow \theta = \frac{\pi}{6}$	A1	AG Correctly shown	Accept substitution of $\theta = \frac{\pi}{6}$ into V' to show that $V' = 0$ for second A mark
		ii	$\frac{d^2V}{d\theta^2} = mga(\sin \theta + \sqrt{3} \cos \theta)$	M1	Attempt at second derivative (or first derivative) test	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	when $\theta = \frac{\pi}{6}, \frac{d^2V}{d\theta^2} = 2mga > 0, \text{ so}$ equilibrium is stable	A1	Correctly value of V'' and > 0 Examiner's Comments The part was very well done although some candidates did not show sufficient detail in establishing the given result. A small minority, while correctly showing the position of equilibrium, did not go on to show that this position was stable.	
		iii	KE of system: $T = \frac{1}{2} \left(\frac{4}{3} ma^2 \right) \dot{\theta}^2$	B1		
		iii	$\frac{2}{3} ma^2 \dot{\theta}^2 + mga(\sqrt{3} - \sin \theta - \sqrt{3} \cos \theta) = E$	M1	Attempt at formulation of relevant energy equation ($T + V = E$)	
		iii	$\frac{4}{3} ma^2 \ddot{\theta} + mga(-\cos \theta \dot{\theta} + \sqrt{3} \sin \theta \dot{\theta}) = 0$	M1	Differentiates their energy equation	Condone absence of θ throughout – but if inconsistent then M0
		iii	$\frac{4}{3} a \ddot{\theta} = g(\cos \theta - \sqrt{3} \sin \theta)$	A1	AG Correctly shown Examiner's Comments The method of differentiating the energy equation to obtain the given equation of motion was quite well understood with the majority of candidates earning full marks for this part.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$\frac{4}{3}a\ddot{\phi} = g \cos\left(\phi + \frac{\pi}{6}\right) - \sqrt{3}g \sin\left(\phi + \frac{\pi}{6}\right)$	M1*	Substituting $\theta = \phi + \frac{\pi}{6}$ and genuine attempt to expand both terms	
		iv	$\frac{4}{3}a\ddot{\phi} = g \left(\cos\phi \cos\frac{\pi}{6} - \sin\phi \sin\frac{\pi}{6} \right) - \sqrt{3}g \left(\sin\phi \cos\frac{\pi}{6} + \sin\frac{\pi}{6} \cos\phi \right)$	A2	Correct trigonometric expansion, $\ddot{\theta} = \ddot{\phi}$ and using both $\sin\frac{\pi}{6} = \frac{1}{2}$, $\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$ (one error A1)	
		iv	$\frac{4}{3}a\ddot{\phi} = g \left(\frac{\sqrt{3}}{2} \cos\phi - \frac{1}{2} \sin\phi \right) - \sqrt{3}g \left(\frac{\sqrt{3}}{2} \sin\phi + \frac{1}{2} \cos\phi \right)$	M1dep*	Apply small angle approximation	
		iv	$\frac{4}{3}a\ddot{\phi} = -2g \sin\phi$	A1	$\ddot{\phi} \approx -\omega^2\phi$ and must state this is (approx.) simple harmonic – condone $\ddot{\theta}$	
		iv	Approximate period is $2\pi\sqrt{\frac{2a}{3g}}$	A1	Cao Examiner's Comments This part was generally well done with the majority of candidates correctly applying the given substitution and going on to show that the motion of the system was approximately simple harmonic. The most common mistakes in this part were due to errors in the expansion of the trigonometric terms, failing to substitute for $\ddot{\theta}$, not applying the small angle approximation and not stating that the motion was (approximately) simple harmonic.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	19		
19			$M = \int_0^a k \left(4 - \sqrt{\frac{x}{a}} \right) dx$ $= k \left[4x - \frac{2}{3} a^{-\frac{1}{2}} x^{\frac{3}{2}} \right]_0^a \quad (= \frac{10}{3} ka)$ $M\bar{x} = \int_0^a k \left(4 - \sqrt{\frac{x}{a}} \right) x dx$ $= k \left[2x^2 - \frac{2}{5} a^{-\frac{1}{2}} x^{\frac{5}{2}} \right]_0^a \quad (= \frac{8}{5} ka^2)$ $\bar{x} = \frac{\frac{8}{5} ka^2}{\frac{10}{3} ka}$ $= \frac{12}{25} a = 0.48a$	M1 A1 M1 A2 M1 A1	For $\int \left(4 - \sqrt{\frac{x}{a}} \right) dx$ For $4x - \frac{2}{3} a^{-\frac{1}{2}} x^{\frac{3}{2}}$ For $\int \left(4 - \sqrt{\frac{x}{a}} \right) x dx$ For $2x^2 - \frac{2}{5} a^{-\frac{1}{2}} x^{\frac{5}{2}}$ Dependent on previous M1M1 <u>Examiner's Comments</u> The techniques for finding the centre of mass of a rod with variable density were very well understood, and were usually carried out accurately. Some candidates made errors with the integration of $\sqrt{x/a}$ and $x\sqrt{x/a}$.	Give A1 for one correct term
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance	
20		i	Moments about A for AB	M1	Allow if sign errors, $\sin\theta$ / $\cos\theta$ present; dim correct	Allow sin/cos confusion.
		i	$Wl\frac{3}{5} + F2l\frac{4}{5} = R2l\frac{3}{5}$	A1	Allow sign errors	
		i	$3W + 8F = 6R$ oe	A1	CAO	Examiner's Comments Almost all candidates followed the instructions and found a correct equation relating W, R and F . A small number of candidates made errors confusing sine and cosine, and some made slips with signs.
		ii	Moments about A for AC	M1	Allow if sign errors, $\sin\theta$ / $\cos\theta$ present	Q is normal reaction at C Allow sin/cos confusion.
		ii	$2W2l\frac{4}{5} + G4l\frac{3}{5} = Q4l\frac{4}{5}$	A1	Allow sign errors ($16W + 12G = 16Q$ oe)	G is friction force at C
		ii	$F = G$	B1		
		ii	Solve with eqn from (i) and $R + Q = 3W$	M1		dep moments equation
		ii	$R = \frac{73}{50}W$	A1	AG	Accept $1.46W$
		ii	$Q = \frac{77}{50}W$	A1	If < 4 marks, sc B1 for $R + Q = W + 2W$	Or $1.54W$
		ii	Or moments about C for whole system	(M1)	Allow if sign errors, $\sin\theta$ / $\cos\theta$ present. Need 4 terms.	Allow sin/cos confusion.
		ii	$2W \times 2l\frac{4}{5} + W \left(4l\frac{4}{5} + l\frac{3}{5} \right) - R \left(4l\frac{4}{5} + 2l\frac{3}{5} \right) = F \left(4l\frac{3}{5} - 2l\frac{4}{5} \right)$	(A1)	Allow sign errors	
		ii		(A1)	Correct	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$(35W - 22R = 4F)$ Solve with eqn from (i)	(M1)	Final A1A1 as main scheme	dep moments equation Examiner's Comments This part could be tackled in a number of ways and most, but not all, candidates attempted to find an appropriate second equation to solve along with their answer to part (i). The most common second equation was found by taking moments about A for the equilibrium of rod AC, and candidates then had to use the fact that the friction forces at B and C were equal together with the fact that the sum of the normal reactions at B and C was $3W$. The next most common second equation was to take moments about C for the equilibrium of the whole system; quite a number of candidates attempting this method had the friction force at B acting in the wrong direction, while others omitted this term. Another method was to take moments about C for the equilibrium of AC; this required candidates to use the forces at A, and many did this quite successfully. Some candidates omitted to find the vertical component of the force acting on AC at C; a small number of candidates, having failed to show the given result, assumed the value of R and then found the vertical component of the force acting on AC at C.

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Attempt to find $F (= G) = \frac{18}{25}W$ or $F (= G) = \frac{36}{73}R$	M1	Or 0.72 W Ft their normal forces if at least M1 in (i) and (ii)	Or use (i) with $F = \mu R$ and $\frac{73}{50}W$ M1A1
		iii	AB	B1	Can be implied	$584\mu W = 438W - 150W$
		iii	Coeff of friction $= \frac{18}{25}W \div \frac{73}{50}W$	M1	Or $0.72W/1.46W$ May see $\frac{18}{25}W \div \frac{77}{50}W$ at C $\left(\frac{36}{77}\right)$	Solve A1; B1 as scheme
		iii	$\frac{36}{73}$ or 0.493(15)	A1	CAO	Examiner's Comments While there were many correct, efficient solutions to this part, not all candidates realised that rod AB would be the one on the point of slipping. Some candidates got over this problem by finding the possible values of the coefficient of friction for both B and then C slipping - and most of these then gained full marks by then indicating rod AB.
			Total	13		



Question			Answer/Indicative content	Marks	Part marks and guidance	
21		i	Moments about B for BC	M1	2 terms involving $\sin\beta$ and $\cos\beta$, 75 and 50	Allow $\sin / \cos$ error / sign error
		i	$75L\cos\beta = 50 \times 2L\sin\beta$	A1		Allow missing L
		i	$\tan\beta = 3/4$	A1	WWW AG Examiner's Comments Very few candidates had any problem at all with this part, though there was a very small number of candidates who made no real attempt at this question.	
		ii	Moments about A for both rods	M1*	All (5) terms present; each term involves $\sin / \cos \alpha/\beta$. Dim correct: no extra terms	Allow $\sin / \cos, L/2L$, sign errors
		ii	$WL\cos\alpha + 75(2L\cos\alpha + L\cos\beta) = 50(2L\sin\alpha + 2L\sin\beta)$	A1		L may be cancelled
		ii	Correct values for $\sin / \cos \alpha/\beta$	B1	dep M1A1	All 4 seen
		ii	Attempt to solve	*M1	dep B1 also	All values substituted
		ii	($W=$) 90 (N)	A1		
		ii	OR 'X' = 50 N to right on AB oe	B1	sc B1 for magnitudes if directions	50 & 75 may be seen on diagram in (i)
		ii	'Y' = 75 N down on AB oe	B1	Wrong / missing	
		ii	Moments about A for AB	M1	Involves $W, 75, 50, \sin \alpha$ and $\cos \alpha$. Dimensionally correct; no extra terms	
		ii	$WL \cos \alpha$ or $+ 75 \times 2L \cos \alpha = 50 \times 2L \sin \alpha$	A1	With substitution for α	L may be cancelled



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	($W=$) 90 (N)	A1	Examiner's Comments	
					About half the candidates took moments about A for the two rods together. Other candidates first worked out the horizontal and vertical components of the force acting on rod AB at B and then took moments about A for rod AB . (A variation was to find the horizontal and vertical components of the force acting on AB at A , and take moments about B). A small number of candidates using the former method made errors in the values of one or more of the trigonometric expressions used, and a small number of candidates using the latter method made errors in the directions of the components.	
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i	Using $F = ma$	M1	Must have the right 3 terms; allow sign error / sin for cos for M1	$v^2 = u^2 - 9.8(1 - \cos\theta)$
		i	$T - 0.2g\cos\theta = 0.2v^2/0.5$	A1		
		i	By energy	M1		
		i	$\frac{1}{2} \times 0.2u^2 = \frac{1}{2} \times 0.2v^2 + 0.2g \times 0.5(1 - \cos\theta)$	A1		
		i	$T = 5.88\cos\theta + 0.4u^2 - 3.92$	A1	AG with no errors and no gaps in argument	
					Examiner's Comments	
					Most candidates had little trouble with this part question, though some did not show sufficiently clearly how they derived the given answer. A small number of candidates, having initially failed to arrive at the given answer, did the question again, or went back and corrected their work.	
		ii	When $\theta = 180^\circ$, $5.88\cos\theta + 0.4u^2 - 3.92 = 0$	M1	Allow inequalities for M1A1	4.9497... Not > 4.95
		ii	$-5.88 + 0.4u^2 - 3.92 = 0$	A1		
		ii	Min u is 4.95 (m s ⁻¹)	A1	$\frac{7}{2}\sqrt{2}$	
		ii	OR, at top, $mg = \frac{mv^2}{r}$, so $v^2 = 0.5g$	B1	Allow inequalities for B1M1	
		ii	By energy $\frac{1}{2} \times 0.2u^2 = \frac{1}{2} \times 0.2 \times 0.5g + 0.2g$	M1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Min u is $4.95 \text{ (m s}^{-1}\text{)}$	A1	Examiner's Comments Quite a number of candidates, including some with high marks, assumed that the particle would be stationary at the highest point, and arrived at the wrong answer of 4.43 m s^{-1} . Other candidates thought the value of θ at the uppermost point was $0, \pi/2$ or 2π	
		iii	$5.88\cos\theta + 0.4 \times 12.25 - 3.92 = 0$	M1		
		iii	$\cos\theta = (3.92 - 4.9)/5.88 (= -1/6)$	A1	Might see $\theta 99.6^\circ$ or 1.74 radians	$99.49406\dots^\circ, 1.73824\dots\text{rads}$
		iii	Use energy eq ⁿ from (i)	M1	Accept use of their θ	
		iii	$\frac{1}{2} \times 0.2 \times 3.5^2 = \frac{1}{2} \times 0.2 v^2 + 0.2g \times 0.5(1 - \cos\theta)$ $v = 0.904 \text{ m s}^{-1}$	A1	$\frac{7}{30}\sqrt{15}$	$0.903696\dots$
		iii	OR use T equation from (i)	M1		
		iii	$0 - 0.2g(-1/6) = 0.2v^2/0.5$ $v = 0.904 \text{ m s}^{-1}$	A1	Examiner's Comments The most common error on this part was to make a sign slip and arrive at $\cos\theta = \frac{1}{6}$. A significant minority of candidates found the angle θ and then stopped before finding the speed.	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
23		i	Moments about C for CD	M1	Examiner's Comments	allow M if sin / cos wrong
					Although the working was often rather complicated, most candidates did manage to show that the force on CD at D is $\frac{1}{2}W$.	
					While most then went on to find the force acting on AB at B a small number missed out this demand and others did very complicated and wrong work.	
		i	$W/\sqrt{3}/2 (\cos 30^\circ) = Q/\sqrt{3} (\cos 30^\circ)$	A1		
		i	$(Q =) W/2$	A1	AG	
		i	Resolve vert	M1		
		i	$(R =) \frac{3}{2}W$	A1	CAO	
		ii	$X = 0$	B1	Examiner's Comments	
					This part was obvious to quite a number of candidates, but a large number of candidates did very complicated and usually wrong work to arrive at quite exotic wrong answers.	
		ii	Resolve vert for CD or AB		$Y + Q = W$ or $Y + W = R$	
		ii	$Y = W/2$	B1*		
		ii	Vertically downwards	*B1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Moments about C for AB	M1	Examiner's Comments This part was found quite challenging – only candidates with the correct answer were able to gain full marks, but other candidates were given method marks for correct working.	Allow M if sin / cos wrong or sign errors; need all terms
		iii	$P\cos 30^\circ + F\cos 30^\circ = R/\sin 30^\circ$	A1	Correct	
		iii	Use P in terms of F	M1	$F = P$ or other correct 2nd step	Allow if missing term above
		iii	Find F in terms of W, or in terms of R	M1	$F = \frac{\sqrt{3}}{4}W$	Or getting 'their' F oe, ie putting $F = \mu R$ in moment equation.
		iii	$\mu = (F/R) = \sqrt{3}/6$	A1	Accept decimal answers from 0.288675	
		iii	OR Moments about A for AB	M1		Allow M if sin / cos wrong or sign errors; need all terms
		iii	$W/\sin 30^\circ + (Y)/\sin 30^\circ + F/2\cos 30^\circ = R/2\sin 30^\circ$	A1		May have X term if not 0 in (ii)
		iii	Write Y (and X) in terms of W	M1		
		iii	Find F in terms of W, or in terms of R, oe	M1	$F = \frac{\sqrt{3}}{4}W$	
		iii	$\mu = (F/R) = \sqrt{3}/6$	A1	Accept decimal answers from 0.288675	
			Total	13		



Question			Answer/Indicative content	Marks	Part marks and guidance	
24		i	Take moments about A for whole body	M1	Correct 3 terms needed; dim correct	Allow sign errors, $W/2W$, $\cos / \sin$,
		i	$W \times 2L\cos 60^\circ + 2W \times 6L\cos 60^\circ = R \times 8L\cos 60^\circ$	A1	$\cos 60^\circ$ may be omitted	R is reaction at C
		i	$R = 1.75 W$	A1	At least 1 correct step to show given answer	S is reaction at A
		i	$S = 1.25 W$	B1	Examiner's Comments Candidates found this statics question easier than those in recent years, though there were still a few who made no meaningful attempt at this question. Most candidates took moments about A , as hinted at in the question, with many of them correctly dispensing with $\cos 60^\circ$, and even L , at an early stage to reach the answer very quickly. A few took moments about B for the whole body to find an equation involving the reactions at both A and C , but generally solved this correctly by using the result of resolving vertically for the whole body. Others first took moments about B for AB and BC separately, thus involving T ; full marks were available for those who correctly established the given result. A small number of candidates gained only 1 mark for the force acting at A .	For less efficient methods, M1 can only be earned when equation with one unknown, R , is reached.
		ii	Take moments about B for equil of BC	M1*	Correct 3 resolved terms needed; dim correct; or for BA	Allow sign errors, $W/2W$, $\cos / \sin$,



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$T \times L \sin 60^\circ + 2W \times 2L \cos 60^\circ = 1.75W \times 4L \cos 60^\circ$	A1	$T \times L \sin 60^\circ + W \times 2L \cos 60^\circ = 1.25W \times 4L \cos 60^\circ$	
		ii	solve to get	*M1		
		ii	$T = \sqrt{3}W$	A1	Accept $T = 1.73W$	
					Examiner's Comments	
					Most candidates were successful here, though some made errors in resolving forces.	
		iii	Resolve vertically for AB $Y + 1.25W - W = 0$	M1		Weight and normal term must be for same rod
		iii	$Y = 0.25W$, downwards	A1CAO	Direction must be clear	
		iii	$X = \sqrt{3}W$ to left	B1ft	Direction must be clear	
					Examiner's Comments	
					This part was usually correct, though sometimes on follow through from part (ii). A small number of candidates got the directions of forces wrong, or else did not make the directions clear.	
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Question			Answer/Indicative content	Marks	Part marks and guidance	
25		i	Moments about B for equilibrium of BC	M1	$2Wl\cos 60^\circ + F2l\sin 60^\circ = R2l\cos 60^\circ$ Examiner's Comments This was usually correct, with almost all candidates managing to give sufficient detail of their working to convince examiners that they were right.	3 moment terms, condone sin / cos errors and missing l . Need trig terms for M1
		i	$W + \sqrt{3} F = R$ AG	A1	Must be formula for R	correct, with sin / cos evaluated
		ii	Moments about A for equilibrium of whole system	M1	At least one of F and R terms must involve lengths of both rods Examiner's Comments Many candidates produced almost exactly the complicated equation shown in the mark scheme, with a few using equivalent trigonometric expressions and just a few making slips with the angles. Most of these then went on to give a correct final answer. A small number of candidates first worked out the components of the force acting on AB at B . Full credit was available for this method though most of those using it made sign errors. A small number of candidates did not appreciate that having two rods meant that the moment terms for F, R and $2W$ had separate length parts within them.	At least 3 moment terms, condone sin / cos errors, sign errors and $l / 2l$ confusion / missing. Wrong use of forces at B gets M0
		ii		A1	$Wl\cos 30 + 2W (2l\cos 30 + l\cos 60) + F (2l\sin 60 + 2l\sin 30) = R (2l\cos 30 + 2l\cos 60)$	4 terms, accept sin / cos errors and $l / 2l$ confusion / missing and sign errors for A1



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	sin / cos left in, but correct	
		ii	$W\left(\frac{5\sqrt{3}}{2}+1\right)+F(\sqrt{3}+1)=R(\sqrt{3}+1)$	A1	fully correct, oe. Mark final answer Allow full credit for candidates who work out internal forces at B and work correctly from there.	accept $5.33W + 2.73F = 2.73R$, $W\left(\frac{13}{4}-\frac{3\sqrt{3}}{4}\right)+F=R$ Eg $3R = \sqrt{F} + 7.5W$
		iii	Solving 2 sim equations to eliminate F or R	M1	Both equations must involve W, F and R Examiner's Comments There were many good concise solutions to this part, and also quite a lot of complicated rambling solutions. In general, the longer the solution the more chance there was of an error creeping in. A significant number of candidates were sufficiently confident to work throughout in surds and leave the answer as $\frac{3\sqrt{3}}{13}$	allow slips in working
		iii		A1	$F = \frac{3\sqrt{3}}{4}W$	$F = 1.299 W$
		iii		A1	$R = \frac{13}{4}W$	$R = 3.25 W$
		iii	Use $F = \mu R$ to find μ	M1	At any point	
		iii	$(\mu =) \frac{3\sqrt{3}}{13} \quad (0.39970)$	A1	Or eliminate W Use $F = \mu R$ cao	Accept 0.4 if with correct working $5.33(R - 1.73F) + 2.73F = 2.73R$ $2.6R = 6.52F$
					M1A1A1 M1 A1	



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	11	



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Question			Answer/Indicative content	Marks	Part marks and guidance	
26		i	CoM of one semicircular $\frac{4a}{3\pi}$ lamina is $\frac{4a}{3\pi}$	B1	oe, may be unsimplified, allow $\frac{4r}{3\pi}$	
		i		M1	Table of values idea	
		i	$-\left(\frac{1}{2}\pi a^2\right)\left(\frac{4a}{3\pi}\right) + \left(\frac{1}{2}\pi(3a)^2\right)\left(\frac{12a}{3\pi}\right)$	A1		
		i	$= \left(\frac{1}{2}\pi(3a)^2 - \frac{1}{2}\pi a^2\right)x_G$	A1		
		i	$x_G = \frac{13a}{3\pi}$	A1	AG Correctly shown Examiner's Comments Although this is a standard question, significant errors were made in an attempt to show the given answer. The key word "lamina" was often missed by candidates, who proceeded to use the centre of mass of an arc rather than a sector. It was also common to see a candidate thinking the shape was a framework. Even more concerning was the number of candidates who used the area of full circles in their calculation, which did lead to the given answer so that they were blissfully unaware of the error that had been made.	
		ii		M1	Moments equation in terms of a and T , with correct number of terms, dimensionally correct, resolving on T side of equation; if conflicting evidence about point taking moments mark to benefit of candidate.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$3g\left(\frac{13a}{3\pi}\right) = (T \cos 40)(6a)$	A1		
		ii	$T = 8.82\text{N}$	A1	$T = 8.822960\dots$ Examiner's Comments Candidates are improving in indicating the point about which they are taking moments and the most obvious place was about point A. However some used AB as 3a in their working instead of 6a, as well as others using $\sin 40^\circ$ in an attempt to find the distance associated with the tension. Some chose to take moments about other points, the most common being O, but proved to be less successful in answering the question. The lack of a full force diagram meant that moments of the force(s) at the pivot were omitted.	
		iii	$X = T \cos 40$	B1	ft candidates value of T if substituted ($X = 6.758779\dots$)	
		iii	$Y = 3g + T \sin 40$	B1	ft candidates value of T if substituted ($Y = 35.071289\dots$)	
		iii	$\tan \theta = \frac{35.0712\dots}{6.7587\dots}$	M1	Any relevant angle	
		iii	$\theta = 79.1$ <u>above</u> horizontal (to the left)	A1	$\theta = 79.0919\dots$ (10.9 to the upward vertical); above or upwards may be implied by a correct diagram.	
		iii	OR			
		iii	$x^2 = (3g)^2 + T^2 - 2 \times 3g \times T \times \cos 130$	M1A1 ft		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Use of sine rule to find either of the missing angles	M1	$\theta = 79.0919\dots$ (10.9 to the upward vertical); above or	
		iii	$\theta = 79.1$ above horizontal (to the left)	A1	upwards may be implied by a correct diagram. Examiner's Comments A good attempt was made at this with candidates gaining credit if they had a wrong tension in part (ii). However a significant number were not awarded the final mark due to the lack of clarity of the direction. Candidates must be aware that 79.1° to the horizontal is insufficient as this could mean either above or below the horizontal. A diagram would have helped to clarify the situation.	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
27		i	$Fr = R_w$	B1	Resolving horizontally	
		i		M1	Moments about A all forces accounted for	
		i	$Wa \cos \theta + 6Wx \cos \theta = 2aR_w \sin \theta$	A1	A1 for at least two terms correct, may involve θ	
		i		A1	Fully correct equation without θ	
		i	$Fr = \frac{5W}{24} \left(1 + \frac{6x}{a} \right)$	A1	AG Correctly shown	
		i	OR $R_g = 7W$	B1	Resolving vertically	
		i	$W \cos \theta + 6W(2a - x) \cos \theta + 2aFr \sin \theta = 2aR_g \cos \theta$	M1	Moments about B all forces accounted for	
		i		A1A1	A1 for two terms correct	
		i	$Fr = \frac{5W}{24} \left(1 + \frac{6x}{a} \right)$	A1	AG Correctly shown	
					Examiner's Comments	
					A good diagram showing all forces acting on the ladder was beneficial to those who included it. Taking moments about A was the most common method to find the friction. However this method does initially lead to the normal reaction at the wall being the given answer. Some went no further thinking they had shown what was required. Those who took moments about B were less successful, mainly due to at least the moment of one force missing, typically the normal reaction at A .	
		ii	$R_g = 7W$	B1	Resolving vertically (maybe seen in (i) and used in (ii))	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{5W}{24} \left(1 + \frac{6x}{a} \right) = \frac{1}{3} (7W)$	M1	Use of $Fr = \mu R$ with candidates R_g ; allow $\leq, <, >, \geq$	
		ii	$x = 1.7a$	A1	17a/10, 51a/30; must be = or $\leq$ Examiner's Comments Many candidates made a reasonable attempt to use $F = \mu R$ after finding the normal reaction on at B, but many couldn't solve this to get x .	
		iii	$\frac{Wa \cos \alpha + 6W(2a) \cos \alpha}{2a \sin \alpha} = \frac{1}{3} (7W)$ (oe for moments about B)	M1	Setting $x = 2a$ in their dimensionally correct moments equation and use of $Fr = \mu R$ allow $\leq, <, >, \geq$	
		iii		A1		
		iii	$\tan \alpha = \frac{39}{14}$	A1	$\tan \alpha = 2.7857142\dots$; must be = or $\geq$ Examiner's Comments This proved to be the most difficult request on the paper. Of those who made any attempt, many, wrongly, assumed that the friction found in part (i) still applied even though this was a completely new scenario. A significant number of candidates did not even attempt the question.	
			Total	11		



Question			Answer/Indicative content	Marks	Part marks and guidance	
28		i	$2(8)\sin(\pi/2)/(3\pi/2)$	B1	CoM of semi-circle (3.395305.....); $(32/3\pi)$	
		i		M1	Table of values idea to get an equation/expression, using any fixed axis	
		i	$(80 + 32\pi) x_G$	A1		
		i	$=80(2) + 32\pi(4 + cv(\text{CoM}))$	A1ft	Relative to the axis they are using; ft their CoM	
		i	$x_G = 5(.00) \text{ cm}$	A1	(5.004444492 cm) Examiner's Comments A significant number of correct solutions were seen to this standard type of question, but there were two common errors seen. Firstly the centre of mass of the semi-circle was seen as 3 cm from use of the wrong formula, $\frac{3r}{8}$, and was also seen as 4 cm. Secondly, a significant number of candidates had an incorrect area for the semi-circle with values of 64π and 8π commonly seen. Some also worked out the distance of the centre of mass from AE here (or in part (ii)) even though this was not required.	
		ii		M1	Moments about E or A; allow wrong distance for either force	
		ii	$4(20) = W(x_G \text{ from (i)} - 4)$	A1ft	ft on $cv(x_G)$; mg OK here; if $x_G < 4$ need $(4 - x_G \text{ from (i)})$	
		ii	$4(20) = W(x_G \text{ from (i)})$	A1ft	ft on $cv(x_G)$; mg OK here	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	(Greatest) $W = 80 \text{ N}$	A1	79.64601393 from exact x_G ; allow anything between 79.6 and 80	
		ii	(Least) $W = 16 \text{ N}$	A1	15.98579026 from exact x_G SC for masses found only, 4/5 for both 8.13 to 8.16 and 1.63 Examiner's Comments This question requested two values, but it was very common to see only one value given. The key points <i>A</i> and <i>E</i> appeared to be missed by many candidates. There were also many candidates who appeared confused between mass and weight, such that it was common to see the final answer as mass.	
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
29		i		*M1	Moments about A, all terms needed and no extra; dimensionally correct; each term must include $\sin\theta$, or $\cos\theta$ or $\tan\theta$	
		i	$2W(1\cos\theta) + W(2\cos\theta) =$	A1		
		i	$R_B(4\sin\theta)$	A1	May have Fr for R_B	
		i	$R_A = 3W$	B1		
		i		dep*M1	Use of $Fr = \mu R_A$ to get an equation in W (or R_A) only	
		i	$\mu = 2/3$	A1	AG; must come from exact θ	
		i	OR	*M1	Moments about B, all terms needed and no extra; dimensionally correct; each	
		i	$W(2\cos\theta) + 2W(3\cos\theta)$	A1	term must include $\sin\theta$, or $\cos\theta$ or $\tan\theta$	
		i	$+ Fr(4\sin\theta) = R_A(4\cos\theta)$	A1		
		i	$R_A = 3W$	B1		
		i		dep*M1	Use of $Fr = \mu R_A$ to get an equation in W (or R_A) only	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$\mu = 2/3$	A1	AG; must come from exact θ Examiner's Comments Some very elegant answers were seen to this part, but they were a rarity. For those who knew what they were doing, this was a standard resolve and take moments request. The usual place to take moments about was A , but a significant number of candidates used the alternative of B . The most common error here was failing to include all the necessary terms when taking moments about B . Other errors included missing out a distance from one or more terms (often the 4 was omitted), putting a vertical force at A , putting the force at A perpendicular to the ladder instead of perpendicular to the wall and using an inexact value for θ instead of dividing through so they could make use of the given value of $\tan\theta$.	
	Copyright © 2024 Exam Papers Practice	ii		*M1	Moments about A , all terms needed and no extra; dimensionally correct; each term must include $\sin\theta$, or $\cos\theta$ or $\tan\theta$	
		ii	$(W + aW)(2\cos\theta) = R_B(4\sin\theta)$	A1	May have Fr for R_B	
		ii	$R_A = 3W + aW$	B1	Resolve vertically	
		ii		dep*M1	$Fr \leq \mu R_A$ to get an inequality in W and a only; allow equality here	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$a < 3$	A1	Allow $a < 3$. If using equality, correct inequality need not be justified.	
		ii	OR For first 2 marks	*M1	Moments about B , all terms needed and no extra; dimensionally correct; each	
		ii	$(W + aW)(2\cos\theta) + 2W(4\cos\theta) + Fr(4\sin\theta) = RA(4\cos\theta)$	A1	term must include $\sin\theta$, or $\cos\theta$ or $\tan\theta$ Examiner's Comments Again some candidates took moments about B , when the easier route was to take moments about A . Invariably those who found part (i) difficult also struggled here, omitting terms from their moments equation or having additional terms from placing the extra weight in the incorrect place. A few managed to reach $a = 3$ but then failed to give their answer as a range of values.	
			Total	11		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
30		i	CoM of triangle = $\frac{1}{3} \times cv(12)$ from BD	B1	OR $\frac{2}{3} \times cv(12)$ from C. CoM of triangle	
		i		M1	Table of values idea	
		i	$(80 + 60)x_G$	A1		
		i	$= 4(80) + 12(60)$	A1		
		i	$x_G = 7.43 \text{ cm}$	A1	7.42857... or $\frac{52}{7} \text{ cm}$	
					Examiner's Comments	
					The process of finding the centre of mass of a combined object was well known and many fully correct answers were seen. However a significant number of candidates found difficulty in dealing with the triangle, either by using Pythagoras incorrectly to find the height, or finding the area of the triangle incorrectly, for which 30 cm ² and 65 cm ² were commonly seen.	
		ii	$\tan\theta = (8 - x_G)/5$	M1	Using tan to find a relevant angle	
		ii	$\tan\theta = 0.5714.../5$	A1ft	ft their x_G to target angle with the vertical	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\theta = 6.52^\circ$	A1	6.5198... Allow 6.5(0) from $x_G = 7.43$ Examiner's Comments The majority of candidates who did not arrive at the correct answer in (i) managed to subtract their value for x_G from 8 and used their answer correctly with tan so gaining 2/3 marks. A minority used 10 instead of 5 for the denominator in the trig equation and a small minority subtracted their value for x_G from 10.	
			Total	8		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
31		i		M1	Moments about P	
		i	$18(10) - T(20\sin\theta) + 3(6) = 0$	A1	Need a value for $\sin\theta$ or θ	
		i	$T = 16.5 \text{ N}$	A1	Exact Examiner's Comments The best approach was to take moments about the point P , which many attempted successfully. However some just resolved the tension and forgetting to multiply by the distance PQ . Some candidates thought, incorrectly, that resolving would be sufficient and so totally ignoring any forces at P .	
		ii	$X = T\cos\theta$	B1ft	ft candidates value of T . Resolve horizontally ($X = 13.2 \text{ N}$) or moments; Need a value for $\cos\theta$ or θ	
		ii		M1	Resolve vertically or moments	
		ii	$Y + T\sin\theta - 18 - 3 = 0$	A1ft	ft candidates value of T . $Y = 11.1 \text{ N}$; Need a value for $\sin\theta$ or θ	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$R = \sqrt{(13.2^2 + 11.1^2)} = 17.2$ N	A1	$R = 17.2467...$ Examiner's Comments The most successful method employed was to find both horizontal and vertical components of the force exerted by the wall on the rod and combine these components. However a significant number of candidates only found the horizontal component, gaining credit for the vertical component in (iii).	
		iii	$\mu = cv(Y)/cv(X) = 11.1/13.2$	M1	Use of $Fr = \mu R$	
		iii	$\mu = 0.841$	A1	$\mu = 0.8409...$; allow $^{37}/_{44}$ Examiner's Comments The formula $F = \mu R$ was well known and most used their values correctly to find the coefficient of friction. However some used the value of R , the resultant, found in part (ii).	
		Total		9		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
32		i	$x_G = (2 \times 2)/\pi$	B1	$x_G = 1.2732....$ May be seen in (ii), mark only once.	
		i		*M1	Take moments about A or B	
		i	$P(\text{or } X) \times 4 = 0.3g \times x_G$	A1ft	$P = 0.9358....$ ft their x_G for this mark.	
		i	$Y = 0.3g$	B1		
		i	Use $R^2 = X^2 + Y^2$ to find R	dep*M1		
		i	$R = 3.09 \text{ N}$	A1		
					Examiner's Comments This question proved a good source of marks for some candidates, but others lost marks because they were confused by the idea of an arc, and preferred to make it into a lamina. It was very common to see candidates only find the horizontal component of the force at the hinge and not consider that there might be some component to prevent the object from falling.	
	Copyright © 2024 Exam Papers Practice	ii		M1	Attempt at moments, force $\times$ distance = $0.3g \times$ distance	
		ii	$p \times 4 =$	A1		
		ii	$0.3g \times (2\sin 30 + x_G \sin 60)$	A1	$0.3g \times 2.1026....$	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$P = 1.55$	A1	1.545453..... Examiner's Comments There were many poor attempts seen. The majority approached this by considering the components of the weight, but, usually, only considered the moment of one component when in fact both components have a moment about the hinge. The less common approach of using the perpendicular distance to the line of action of the weight also proved to be demanding.	
			Total	10		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
33		i		M1	Table of moments idea. Moments about other axes acceptable	
		i	$4.4x_G = 4 \times \frac{1}{4} \times 8$	A1		
		i	$-0.4 \times \frac{1}{3} \times 10$	A1		
		i	$x_G = 1.52 \text{ cm}$	A1	Allow $^{50}_{33}$ Examiner's Comments This problem on centre of mass was answered extremely well by many candidates and was an excellent source of marks for many. Saw a few errors mainly due to inconsistencies when taking moments about a particular axis. Another common error was to treat the shell as a uniform solid. Some decided to take the volume of the shapes into consideration even though the mass of the individual components were given in the question.	
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		ii		M1		
		ii	$T_{\text{shell}} \times 18 = 4.4g \times (8 - 1.52) \text{ or } T_{\text{cone}} \times 18 = 4.4g \times (10 + 1.52)$	A1ft	Or any other correct moment equation. ft on x_G from (i)	
		ii	$T_{\text{shell}} + T_{\text{cone}} = 4.4g$	M1	May use a second moments equation	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$T_{\text{shell}} = 15.5$ and $T_{\text{cone}} = 27.6$	A1	For both Examiner's Comments This problem on centre of mass was answered extremely well by many candidates and was an excellent source of marks for many. Was again answered well with the most common errors being the omission of g from moments/resolution equations or incorrectly using the distance found.	
			Total	8		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
34		i	$(2 \times 3\sin(\pi/2))/(3\pi/2)$ or equivalent	B1	Centre of mass of semicircle; $4/\pi$	
		i		M1	Table of moments idea about any	
		i	3×6^2	A1	axis.	
		i	$-(\pi \times 3^2/2) \times (6 - 4/\pi)$	A1		
		i	$= (6^2 - \pi \times 3^2/2)x_G$	A1		
		i	$x_G = 1.88 \text{ cm}$	A1		
					Examiner's Comments	
					The majority of candidates used a correct approach for this question. It was surprising the number who did not find the centre of mass correctly from using the wrong formula, using a radius of 6 cm, or using an angle of 90° instead of $\pi/2$. Other causes of lost marks were finding the area of a semicircle incorrectly and adding rather than finding the difference of the area of the square and semicircle to find the total area.	
		ii	$\tan\theta = 1.88/3$	M1	Attempt at a relevant angle	
		ii	$\theta = 32.1^\circ$	A1ft	allow $180-\theta$ & radians (0.561 or 0.560)	
					Examiner's Comments	
					Most scored well in this part as candidates used their wrong value from part (i) correctly without penalty.	
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
35		i	Use of moments	M1	Trig with 3g, no trig with R unless	
		i	$2.5R = 3g \cos 60^\circ \times 2$	A1	using 2 components.	
		i	$R = 11.76 \text{ N}$	A1	Allow 11.8	
					Examiner's Comments	
					Candidates should be advised that the most profitable way of attempting this type of question is to take moments once and resolve twice. It is of course possible to have more than one moments equation, but candidates who do this are usually not as successful. Most candidates approached this by taking moments about A. Most problems arose from having the force at P in the wrong direction, or finding wrong distances.	
		ii		M1	Resolve vertically, 3 terms, comp	
		ii	$R' + R \cos 60^\circ = 3g$	A1ft	(R).	
		ii	$F = R \cos 30^\circ$	B1ft	Using $\cos(R)$	
		ii	Use $F = \mu R'$	M1	Using $\cos(R)$	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\mu = 0.433$	A1	Not $R' = 3g$ for method Allow 0.435 from use of $R = 11.8$ Examiner's Comments Resolving in two directions was by far the more successful approach. Those who tried to take moments, although a valid method, were generally less successful. Examiners reported that the use of normal reaction as weight was seen as often as in previous series.	
			Total	8		